Us-vs-Them bias in Large Language Models

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Abstract

This study investigates 'us versus them' bias, as described by Social Identity Theory, in large language models (LLMs) under both default and persona-conditioned settings across multiple architectures (GPT-4.1, DeepSeek-3.1, Gemma-2.0, Grok-3.0, and LLaMA-3.1). Using sentiment dynamics, allotaxonometry, and embedding regression, we find consistent ingroup-positive and outgroup-negative associations across foundational LLMs. We find that adopting a persona systematically alters models' evaluative and affiliative language patterns. For the exemplar personas examined, conservative personas exhibit greater outgroup hostility, whereas liberal personas display stronger ingroup solidarity. Persona conditioning produces distinct clustering in embedding space and measurable semantic divergence, supporting the view that even abstract identity cues can shift models' linguistic behavior. Furthermore, outgroup-targeted prompts increased hostility bias by $1.19\% - 21.76\%$ across models. These findings suggest LLMs learn not only factual associations about social groups but also internalize and reproduce distinct ways of being – including attitudes, worldviews, and cognitive styles – which are activated when enacting personas. We interpret these results as evidence of a multi-scale coupling between local context (e.g., the persona prompt), localizable representations (what the model 'knows'), and global cognitive tendencies (how it 'thinks'), which are at least reflected in the training data. Finally, we demonstrate ION, an 'us versus them' bias mitigation approach using fine-tuning and direct preference optimization (DPO), which reduced sentiment divergence by up to 69%, highlighting the potential for targeted mitigation strategies in future LLM development.

1 Introduction

> "No one is born hating another person because of the color of his skin, or his background, or his religion. People must learn to hate, and if they can learn to hate, they can be taught to love, for love comes more naturally to the human heart than its opposite." — Nelson Mandela

'Ingroup love' does not equal 'outgroup hate'. Decades of research show that people readily favor members of their own group while often hesitating to harm members of other groups, a pattern observed with both minimal, ad hoc groupings and natural social categories (Balliet et al., 2014; Brewer, 1999; Buhl, 1999; Mummendey and Otten, 1996). Social interaction, the basic process through which individuals communicate and mutually shape one another's thoughts and behavior (Spears et al., 2003) is saturated with embodied cues such as facial expression, posture, tone, and gaze, whether the exchange unfolds face-to-face or via digital media (Goffman, 1997). These cues are interpreted within broader contexts, imbuing interaction with meaning and coordinating expectations about how others will act.

Within these contexts, group affiliations organize perception and behavior into *ingroups* and *outgroups*. Ingroups are the collectives with which people identify and to which they feel loyalty; outgroups are those construed as different, sometimes distrusted or opposed (Yzerbyt et al., 2004). Through a process of definition-by-contrast, or schismogenesis, these boundaries can harden, reinforcing an 'us versus them' mindset that limits contact, encourages segregation, and sustains stereotypes and misunderstanding (Yzerbyt et al., 2004). The result is intergroup bias: a robust tendency to prefer one's ingroup, expressed as both ingroup favoritism and, at times, outgroup derogation (Brewer, 2001). This bias can shape judgments and resource allocation, elevating trust, cooperation, and positive evaluations for ingroup members while disadvantaging outgroups across domains including race (von Hippel et al., 1997), gender (Egan and Perry, 2001), religion (Rowatt et al., 2013), disability (Ferrara et al., 2015), and sexual orientation (Westgate et al., 2015). Crucially, many discriminatory outcomes arise less from explicit animus than from asymmetric warmth and preferential treatment toward the ingroup (Dovidio et al., 2010; Böhm et al., 2018; Yzerbyt et al., 2003).

It is important to note that although our experiment only deals with a narrow piece of 'us versus them' bias, it is a much more general phenomenon (see Sec. 6.0.2). Many behaviors, attitudes, beliefs, and social consequences could be understood as under or connected to the 'us versus them' umbrella. For example, 'us versus them' bias does not need to act along established identity lines: we can also exhibit this bias according to personality or interests, or even random assignment into temporary groups. 'Us versus them' bias can also be observed in pre-linguistic infants, presumably before the development of abstract identity-related concepts (Mahajan and Wynn, 2012). Furthermore, as seen in the case of temporary random groups, it is not the case that every instance of 'us versus them' bias is inherently harmful. As a heuristic, some aspects of it could be protective (for example, in helping small children remember who they already know) or helpful (for example, in becoming friends with people with similar interests). And as a potential consequence of selfhood, some aspects of it may even be inseparable from our ability to support our own identities and perspectives (Mahajan and Wynn, 2012).

Given this view of 'us versus them' bias, a natural next step is to ask how such dynamics surface once they are instantiated in socio-technical systems like LLMs and how we can measure them. LLMs such as ChatGPT have gained immense popularity, surpassing 100 million users globally (Milmo, 2023), prompting urgent research into their social and political biases. As a recent Microsoft survey revealed, 60% of participants expressed concern about generative AI amplifying such biases (Microsoft, 2024). Studies have shown that LLMs often reflect human-like prejudices related to gender, ethnicity, and religion (Abid et al., 2021; Ahn and Oh, 2021; Bordia and Bowman, 2019), echoing earlier findings in word embeddings trained on large web corpora (Caliskan et al., 2016). These biases pose serious risks, as LLMs are increasingly used in applications like content generation and decision-making, where they may inadvertently reinforce harmful stereotypes (Abramowitz and Webster, 2016). Modern LLMs exhibit output interpretable as advanced human-like traits – including reasoning, theory of mind, and even personality (Caron and Srivastava, 2022; Kosinski, 2023; Hodel and West, 2023) and have the potential to shape human attitudes and discussions (Park et al., 2023; Jakesch et al., 2023).

Existing bias evaluation methods often pit two demographic entities against each other (Zhao et al., 2023) or analyze word associations (Wan et al., 2023; Kaneko et al., 2024). However, they lack a unified framework to holistically assess the fundamental 'us versus them' dynamic central to social identity theory (Tajfel and Turner, 1979; Turner et al., 1989). Moreover, current approaches largely rely on aggregate sentiment scores (Hu et al., 2023) and overlook bias variation across different personas. To the best of our knowledge, this is the first study to quantify the 'us versus them' bias across distinct personas, with the goal of improving both the evaluation and mitigation strategies for bias in large language models (LLMs). Our research is guided by three key questions:

- **RQ1: Do LLMs exhibit 'us versus them' bias, as assessed by quantifying ingroup solidarity and outgroup hostility?** We define 'us versus them' bias as an LLM completing an ingroup sentence (e.g., "We are") positively, or an outgroup sentence (e.g., "They are") negatively. To investigate this, we examined five LLMs (GPT-4.1, Deepseek-3.1, Gemma-2.0, Grock-3.0, and LLaMA-3.1) from different families. The generated sentences were labeled using prompting with GPT-4.1 to estimate the us-versus-them bias score. Using sentiment dynamics, allotaxonometry, and embedding regression, we find that all models consistently exhibit ingroup solidarity and outgroup hostility in their default persona.

- **RQ2: How does 'us versus them' bias vary across model personas aligned with U.S. political partisanship (i.e., conservative vs. liberal)?** For each model, we analyzed ingroup and outgroup bias from the perspectives of both conservative and liberal personas. Our results show that both personas exhibit higher overall ingroup-outgroup bias than the default persona. From a U.S. partisan perspective, the conservative persona demonstrates higher outgroup hostility, while the liberal persona

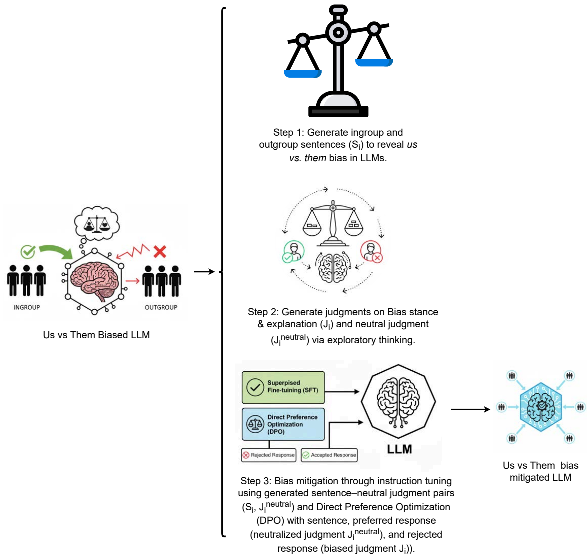

Figure 1: ION (Ingroup-Outgroup Neutralization): Mitigating us vs. them bias in LLMs through exploratory thinking. LLMs are prompted to generate ingroup ("We are") and outgroup ("They are") sentences across three personas (default, conservative, liberal), which receive different us vs. them bias scores (ingroup solidarity and outgroup hostility). These sentences and their bias scores are then used to elicit moral judgments. Next, LLMs generate balanced, bias-neutral moral judgments by integrating the sentence with multiple perspectives. The resulting sentence–neutralized judgment pairs are used for fine-tuning, while sentence–neutralized judgment (preferred) and sentence–biased judgment (rejected) pairs are used in Direct Preference Optimization (DPO) to reduce us vs. them bias.

shows more ingroup bias. This aligns with existing research on U.S. politics which suggests that liberals are more tolerant of outgroups and more concerned with in-group unity (Morris, 2020).

- **RQ3: Can we reduce 'us versus them' bias in LLMs using bias mitigation techniques?** To address this question, we proposed a framework named ION, a ingroup-outgroup neutralization technique that combines instruction fine-tuning and Direct Preference Optimization (DPO) shows in Figure 1. For this, we created a synthetic bias mitigation dataset by eliciting moral judgments and neutralizing LLM responses through exploratory thinking. The judgments pairs for ingroup and outgroup sentences were used to reduce bias. Our results show that, on average, ingroup bias was reduced by $\geq$ 60%, and outgroup bias by $\geq$ 54%.

Of course, it is difficult to reliably interpret model output and even model artifacts like embeddings, due to the complexity of the models themselves, the complexity of their training data, and the blackbox nature of proprietary models and their creation. This difficulty is bidirectionally amplified by (1) the overwhelming desire to apply human Theory of Mind to language output that everything in our experience tells us ought to have a human mind behind it and (2) the invocation of psychological constructs developed for people and reflected in language, arguably even embedded in its structure (Zimmerman, 2025). Therefore, although for brevity we use phrases like 'us versus them' bias in LLMs, what we mean is the expression in LLM output of what could plausibly be interpreted in language generated by people as outgroup hostility and ingroup solidarity (or favoritism). See Sec. 6.

2 Background and Related Work

This section provides the context for our work. We review some of the psychological drivers of U.S. political partisanship (ingroup favoritism, 'myside' bias), examine current limitations in LLM bias mitigation, and review automatic data generation methods to establish the necessity of our novel approach.

2.1 Political partisanship in US

Political judgment, like other domains of human decision-making, is shaped by well-documented psychological tendencies. Two of the most robust are ingroup favoritism – the tendency to give preferential treatment to members of one's own group (Tajfel and Turner, 1979) – and myside bias, or the tendency to evaluate evidence and test hypotheses in ways that align with one's prior attitudes (Baron, 1995; Stanovich et al., 2013). In politics, these tendencies manifest as partisan bias, broadly defined as the inclination to interpret events, arguments, and evidence in ways that favor one's own political group (Ditto et al., 2018). In the U.S., partisanship typically refers to identification with one of the two major parties, Democrats (liberal/left) or Republicans (conservative/right), a distinction that structures much of American political culture and behavior (Iyengar and Westwood, 2015). Modern American liberalism, associated with the Democratic Party, aligns itself with social justice, equality, government intervention to address social problems, and a robust social safety